

torch.signal.windows.exponential#

[https://pytorch.org/docs/stable/generated/torch.signal.windows.exponential.h](https://pytorch.org/docs/stable/generated/torch.signal.windows.exponential.html)

Description:

Computes a window with an exponential waveform. Also known as Poisson window. The exponential window is defined as follows: where c is the center of the window. The window is normalized to 1 (maximum value is 1). However, the 1 doesn't appear if M is even and `sym` is `True`.
 M (int) – the length of the window. In other words, the number of points of the returned window.

Function Signature

```
torch.signal.windows.exponential(M, *, center=None, tau=1.0,
sym=True, dtype=None, layout=torch.strided, device=None,
requires_grad=False)[source]#
```

Parameters

Keyword Arguments

Return type

Parameters

Keyword

center (float, optional) – where the center of the window will be located. Default: $M / 2$ if sym is False, else $(M - 1) / 2$. tau (float, optional) – the decay value. Tau is generally associated with a percentage, that means, that the value should vary within the interval (0, 100]. If tau is 100, it is considered the uniform window. Default: 1.0. sym (bool, optional) – If False, returns a periodic window suitable for use in spectral analysis. If True, returns a symmetric window suitable for use in filter design. Default: True. dtype (torch.dtype, optional) – the desired data type of returned tensor. Default: if None, uses a global default (see torch.set_default_dtype()). layout (torch.layout, optional) – the desired layout of returned Tensor. Default: torch.strided. device (torch.device, op

Returns:

Tensor

Code Examples

```
>>> # Generates a symmetric exponential window of size 10 and
with a decay value of 1.0.
>>> # The center will be at (M - 1) / 2, where M is 10.
>>> torch.signal.windows.exponential(10)
tensor([0.0111, 0.0302, 0.0821, 0.2231, 0.6065, 0.6065, 0.2231,
0.0821, 0.0302, 0.0111])

>>> # Generates a periodic exponential window and decay factor
equal to .5
>>> torch.signal.windows.exponential(10, sym=False, tau=.5)
tensor([4.5400e-05, 3.3546e-04, 2.4788e-03, 1.8316e-02, 1.3534e-
01, 1.0000e+00, 1.3534e-01, 1.8316e-02, 2.4788e-03, 3.3546e-04])
```

Detailed Documentation

Documentation

Computes a window with an exponential waveform. Also known as Poisson window.

The exponential window is defined as follows:

where c is the center of the window.

The window is normalized to 1 (maximum value is 1). However, the 1 doesn't appear if M is even and sym is True.

M (int) – the length of the window. In other words, the number of points of the returned

window.

center (float, optional) – where the center of the window will be located. Default: $M / 2$ if sym is False, else $(M - 1) / 2$.

tau (float, optional) – the decay value. Tau is generally associated with a percentage, that means, that the value should vary within the interval (0, 100]. If tau is 100, it is considered the uniform window. Default: 1.0.

sym (bool, optional) – If False, returns a periodic window suitable for use in spectral analysis. If True, returns a symmetric window suitable for use in filter design. Default: True.

dtype (torch.dtype, optional) – the desired data type of returned tensor. Default: if None, uses a global default (see torch.set_default_dtype()).

layout (torch.layout, optional) – the desired layout of returned Tensor. Default: torch.strided.

device (torch.device, optional) – the desired device of returned tensor. Default: if None, uses the current device for the default tensor type (see torch.set_default_device()). device will be the CPU for CPU tensor types and the current CUDA device for CUDA tensor types.

requires_grad (bool, optional) – If autograd should record operations on the returned tensor. Default: False.